MIL-W-80232B
26 February 1987
SUPERSEDING
MIL-W-80232A
29 October 1979

MILITARY SPECIFICATION

WELDING MACHINE, ARC, AC/DC, TRANSFORMER RECTIFIER, CONSTANT CURRENT, PROGRAMMED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers alternating current (AC) and direct current (DC) transformer-rectifier, constant current welding machines with high-frequency circuit, with gas and water controls and with or without pulsation and slope control.

1.2 Classification. The welding machine shall be of the following types and sizes. The type and size required shall be as specified (see 6.2.1).

Type I - Welding machine with controls for manual welding (see 3.4.4.1).
Type II - Welding machine with controls for manual and semi-automatic welding, with up and down slope control (see 3.4.4.2).
Type III - Welding machine with controls for manual welding with pulsation control (see 3.4.4.3).
Type IV - Welding machine with controls for manual and semi-automatic welding with pulsation control, and with up and down slope control (see 3.4.4.4).

Size 300 - 300 amperes
Size 500 - 500 amperes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3431
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA EW 1 - Electric Arc-Welding Apparatus.

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NEMA ICS 1 - Industrial Control and Systems.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103).

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The welding machine shall be a single-operator combination AC/DC, transformer-rectifier, constant current type. The machine shall be of solid state design and rated at 60 percent duty cycle. The size 300 machine shall have not less than two control ranges for the AC and DC welding current. The low range shall control the welding current from not more

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

than 6 amperes through not less than 30 amperes and one or more additional control ranges shall control the current from not more than 15 amperes through the maximum machine output. The size 500 machine shall control the welding current from not more than 25 amperes through maximum output. The operating arc voltage of each welding machine shall be in accordance with NEMA Standards EW-1, for class I machines. The machine shall be equipped with the type controls specified and the controls shall be interchangeable with each other. The welding machine shall be designed and rated to conform with the requirements of NEMA Standard EW-1. The machine furnished under this specification shall be new and of the manufacturer's current design.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the welding machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials provided such materials will not jeopardize the welding machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the welding machine.

3.2.3 Portability. The welding machine shall be portable and shall be base mounted. Running gear with mounting casters, if required (see 6.2.1), shall be specified by the procuring activity. The welding machine shall be provided with lugs or lifting eyes or other suitable means for handling by crane or hoist. Each lifting eye or lug shall be capable of supporting two and one-half times the weight of the machine. The machine shall also be designed to be lifted by forklift truck. The height dimension of the machine shall not exceed 60 inches.

3.2.4 Energy efficiency. The welding machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding machine that present safety hazards. The safety devices shall not interfere with the operation of the welding machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All welding machine parts, components, mechanisms, and assemblies furnished on the welding machine, whether or not specifically required herein, shall comply with all of the requirement of OSHA 29 CFR 1910 that are applicable to the welding machine itself.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

3.2.6 Mercury restriction. The welding machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the welding machine.

3.2.8 Environmental protection. The welding machine shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the welding machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.2.10 Power supply. Unless otherwise specified (see 6.2.1), the welding machine shall be designed to operate from a nominal 200/230/460-volts, 60-hertz (Hz), single-phase power supply and initially wired to operate on 460 volts.

3.2.10.1 Voltage compensation. Unless otherwise specified (see 6.2.1), each welding machine shall compensate for line voltage fluctuation of ±10 percent and maintain virtually constant welding current output as set by the welder current controls.

3.2.10.2 Open circuit voltage. The open circuit voltage shall be not greater than 80 volts in accordance with NEMA EW-1 when the unit is energized at rated primary voltage.

3.2.11 Temperature rise. The maximum allowable temperature rise of various insulated parts of the machine shall not exceed the temperature limits of NEMA EW-1 for the class insulation system used.

3.2.12 Solid-state components. Solid-state design shall be used throughout for electronic components.

3.3 Construction. The welding machine shall be constructed of parts which are new, without defects, and free of repairs.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the welding machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.4. Components. The welding machine shall consist of, but not be limited to, the following components: enclosure, transformer, rectifier, controls, and ventilation system. The enclosure and control system shall conform with NEMA ICS-1.

3.4.1 Enclosure. The sheet metal enclosure shall be dripproof, so designed and constructed that liquid or solid particle falling on the enclosure from above, at angles up to 15 degrees from the vertical, shall not interfere with the satisfactory operation of equipment. Inlet and outlet electrical connections shall be clearly and permanently marked. The output welding terminals shall be of the stud and hex-nut type and the studs shall be not less than 1/2-inch diameter. Receptacles of equivalent capacity shall be acceptable provided a pair of matching plugs suitable for attachment to the welding leads are included. The welding power terminals shall be recessed or otherwise protected against accidental contact by personnel. All intake and exhaust openings shall be screened or shall be small enough to prevent entrance of trash or rodents.

3.4.2 Transformer. The transformer shall be of the laminated core or shell type. The secondary circuit shall be thoroughly insulated from the primary. Insulation of primary and secondary transformer coils shall be class B, F, or H insulation system in accordance with the requirement of NEMA EW-1.

3.4.2.1 Overtemperature protection. Transformer coil windings shall be protected by a suitable thermal device or devices. The device shall operate to either remove the machine from the line or relieve it of its load to

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

prevent the transformer from reaching injurious temperature. A combination overtemperature and overload device may be used in lieu of separate controls for overtemperature and overload.

3.4.3 Rectifier. The rectifier shall be of the full wave semi-conductor type, of adequate size and design for the intended service. The assembly employed shall be of a design which has had a satisfactory history of field service in arc welding. The operating temperature of the rectifier under maximum load conditions shall not exceed a value which will result in damage to the rectifier or adversely affect performance of the power supply.

3.4.4 Controls. The control panel shall be located on the front of the machine enclosure and all instruments, timing devices, outlets and panel. The control panel shall be recessed, and all electrical terminals except welding terminals shall be contained within the enclosure. Each panel shall be provided with switches for power on/off, for selecting AC, straight or reverse polarity DC, and welding current ranges. A potentiometer shall be of the 10-turn type. Operation of adjustment controls shall be free from binding, and controls shall be of such material and design that a fixed setting will remain after extended exposure to adverse atmosphere conditions. Unless otherwise specified (see 6.2.1), instruments to indicate voltage and amperage shall be furnished. Additional control requirements shall be in accordance with 3.4.4.1 through 3.4.4.4 for the type control specified.

3.4.4.1 Type I controls. Welding machines with the type I controls shall have not less than the following control functions:

a. Arc spot on/off switch with timer (0-5 seconds).
b. Hot start control.
c. Pre-purge (0-15 seconds), post gas flow (.6-60 seconds).

3.4.4.2 Type II controls Welding machines with the type II controls shall have not less than the following control functions.

a. Arc spot on/off switch with timer (0-5 seconds).
b. Hot start control.
c. Pre-purge (0-15 seconds), post gas flow (.6-60 seconds).
d. Starting and stopping welding current control.
e. Up and down slope rates (0-10 seconds).
f. Post heat time (0-25 seconds).
g. Welding start and stop.
h. Weld timer (0-100 seconds).

3.4.4.3 Type III controls. Welding machines, with the type III controls shall have not less than the following control functions:

a. Arc spot on/off switch with timer (0-3 seconds).
b. Hot start control.
c. Pre-purge (2-6 seconds), post gas flow (6-60 seconds).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

d. Pulse/second control (percent of on-time 5-95 percent).
e. Low pulse current adjustable 1-100 percent of high pulse welding current (background setting shall be independent from peak setting).

3.4.4.4 Type IV controls. Welding machines with the type IV controls shall have all the control functions as specified for types I, II, and III and the following additional control devices:

a. Slope on/off control.
b. When specified (see 6.2.1) pulsation delay (0-10 seconds).

Requirements for timing devices of all controls are not less than those shown.

3.4.4.5 Remote control. When required, a remote control shall be furnished with not less than 20 feet of electrical cable, control cable, hose and terminal plugs for connection to the machine. The remote control shall be one or more of the following control functions as specified (see 6.2.1).

a. Foot control with rheostate.
b. Remote control pendant with contactor control switch and current control.
c. Torch with switch and current control with slope control switch.

The foot control shall be compatible to all types of controls. The hand control pendant shall be operable with all control systems. The torch with slope control switch shall initiate the start current upslope to welding level and initiate current downslope control. When specified (see 6.2.1), additional lengths of control cables and hose shall be furnished.

3.4.4.6 Overload protection control. The welding machine shall have a device to remove the machine from the line in the event of a sustained overload or power failure. The overload reset or re-energizing of the welding machine shall be manual.

3.4.4.7 Stabilization control. The electrical system shall have means to minimize erratic changes in output welding current and the machine shall provide a constant current welding power supply within $\pm$ 2 percent of weld current setting.

3.4.5 High frequency unit. A high frequency unit shall be contained in the welding machine. A switch(s) shall be provided to select high frequency energy for starting, or for continuous use during welding operation, or to make the high frequency provision inoperative entirely. An intensity adjustment of high frequency shall be provided, and shall be located in the welding machine. The design of the unit shall incorporate all necessary devices to block high frequency feed back to the welding machine, and to retain the rated capacity and the performance characteristics during all operations for which it is intended. The high frequency unit shall be able to establish a welding arc at 25 feet from the switch with a 1/16-inch

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

gap between the workpiece and the electrode when operated in an inert gas atmosphere.

3.4.6 Ventilation system. Adequate ventilation and cooling of all components to meet the duty cycle specified herein shall be provided by forced air. All openings shall be so located and designed that water cannot be drawn into the machine by the intake air. The cooling fan(s) shall have totally enclosed motor(s) with permanently lubricated bearings. The fan assembly shall be securely fastened to a rigid mount in such a manner to withstand vibration encountered in this type of equipment. Fan blades shall be of material and design to prevent fatigue failure. The motor shall conform with NEMA MG-1.

3.4.7 Gas shielding and water cooling system. A gas shielding system shall be provided to control the flow of the shielding gas through the welding machine and base assembly. The welding machine shall be fitted with inlet and outlet gas and water connections conforming to FED-STD H28. A preflow and post flow timer shall be provided to control the time of flow of gas and water thru the welding machine.

3.5 Welding performance. The welding machine shall perform to applicable requirements of NEMA EW-1. When operated by a skilled welder, the welding machine shall be a satisfactory power source for the following welding operations:

a. AC welding with shielded metal arc process
b. AC welding with gas-tungsten arc process
c. DC welding with straight and reverse polarity wherein a drooping characteristic is required

3.6 Diagram of connections A diagram of input power connections shall be attached to readily accessible inside surface of the welding machine. The diagram shall be protected by a coating of transparent varnish or other suitable material.

3.7 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be applied directly onto the item by environmentally protected decalcomania transfer, metal wraparound tag, stencil, silk screen, or by any other method suitable for the use intended which shall be as permanent as the normal life expectancy of the item to which it is applied and capable of withstanding the environmental test and cleaning procedures specified for the item.

3.8 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each welding machine and contain the information listed below. If the welding machine is a special model, the model designation shall include the model of the basic standard welding machine and a suffix indentified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.9 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.10 Workmanship. Workmanship of the welding machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination or all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. Failure of the item to pass the examination, the tests, or the inspections shall be cause for rejection.

4.6 Examination. The welding machine shall be examined to determine compliance with all requirements of this specification including compliance with all referenced standards.

4.7 Tests.

4.7.1 Rough handling test. The machine shall have a chain or cable attached to its lifting eyes or lugs. The welding machine shall be raised and then dropped to fall freely a distance between 6 inches minimum and 10 inches maximum upon a solid concrete slab. This drop shall be repeated so that impact is made on each of three lower edges of the machine. There shall be no damage to the welding machine as a result of this test other than minor denting of the cited lower edges. This test shall be performed prior to tests of 4.7.2 through 4.7.7.

4.7.2 High potential tests Each welding machine shall be subjected to high potential tests described in "Testing Standard" of NEMA EW-1.

4.7.3 Open circuit voltage test. The welding machine shall be connected to primary voltage as specified in 3.2.10. The open circuit voltage shall be in accordance with 3.2.10.2.

4.7.4 Temperature rise test. With machine at room temperature, the maximum of the primary and secondary windings of the transformer shall be recorded. The welding machine shall be operated at rated load and rated duty cycle until stable temperatures are obtained. Temperatures of the windings shall be recorded at 10 minute intervals until stable temperatures are obtained on three successive readings. The load tests are to be run

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

with resistance loading. Temperature rise of the rectifier assemblies, transformer and various parts shall not exceed the temperature limits specified in NEMA EW-1.

4.7.5 Short circuit and capacity test. The welding machine shall be subjected to a sustained short circuit connected through an external resistance load, when set for maximum current output for 1 minute without indication of damage affecting machine operations. The overload protection device specified in 3.4.4.6 shall be out of the circuit when this test is performed. The machine shall then be tested for capacity output. The machine shall be connected to a load bank, and checked for the proper secondary output on AC and DC settings.

4.7.6 Overload protection test. The welding machine shall be adjusted for maximum output and energized. The overload protection device with thermostatic protection temporarily disconnected, shall operate to either remove the machine from the line or relieve it of its load on application of a dead short to the output terminals. No observable or measureable damage to the machine shall result.

4.7.7 Welding performance test. The following welding test shall be performed by a skilled welder. There shall be no difficulty in establishing and maintaining an arc during the performance test. The welding test shall be performed at a distance of not less than 25 feet from the welding machine. Welding shall be performed using DC straight and reverse polarity and AC current. Test shall be performed using both argon and helium shielding gases. There shall be no substantial change in welding current, when changing arc length from a short arc up to 1/2-inch arc. Welds shall be made using electrodes of AWS classification E-308-16 and E-7018 using both DC reverse polarity and AC, and E-6012 using straight polarity. The high frequency circuit shall be used to initiate the arc on AC and DC and to maintain the arc on AC. A completely penetrating and sound weld, using AC welding current and argon gas, shall be made on 1/32-inch thick aluminum without a backup. No burn through shall develop throughout the weld, however penetration visibility on the back side of the plate is permissible. At least 6 additional sound welds shall be made on 1/2-inch aluminum plate, using all controls and systems of the machine. Each weld shall be at least 6 inches long. The completed welds shall be visually inspected for uniformity, porosity, and penetration.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The welding machine covered by this specification is intended for use with AC welding and DC welding in both straight and reverse polarity for joining all ferrous and non ferrous alloys that are subject to being arc welded.

6.1.1 Type I. The welding machine with a type I control is intended for use with the gas metal arc process, the manual gas tungsten arc process with or without a remote control, and the shielded metal arc process.

6.1.2 Type II. The welding machine with a type II control is intended for use with processes as described in 6.1.1 and a scratch-start Tungsten Inert Gas shielded (TIG) process for welding over 100 feet from the welding machine. The upslope will be used to prevent tungsten spitting. The downslope will be used to crater out the welds.

6.1.3 Type III. The welding machine with type III controls is intended for use with processes as described in 6.1.1 and processes where in pulsation is required.

6.1.4 Type IV. The welding machine with type IV controls is intended for use with all processes as described in 6.1.1, 6.1.2 and 6.1.3.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, and size required (see 1.2).

c. First article, if required (see 3.1).

d. If welding machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. Portability, as specified (see 3.2.3).

f. Power supply, if different specify voltage, phase and Hz (see 3.2.10).

g. Voltage compensation, if not required (see 3.2.10.1).

h. Painting, if different (see 3.3.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

i. Meters, if different (see 3.4.4).

j. Pulsation delay control, if required (see 3.4.4.4).

k. Remote control, if required (specify control device required and specify if ten turn potentiometer on the torch is required (see 3.4.4.5).

l. Nameplate, if different (see 3.8).

m. Inspection conditions, if different (see 4.3).

n. First article inspection, if different (see 4.4).

o. Quality conformance inspection, if different (see 4.5).

p. If preservation, packing, and marking in accordance with MIL-W-45562 are required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing. Welding machine, arc/DC, transformer-rectifier, constant current, programmed:

Control, overload protection, stabilization
Protection, overtemperature
Rectifier
Unit, high frequency

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA - IP

Project (3431-0180)

Review activities:
Navy - AS, YD
Air Force - 84

User activities:
Army - ME
Navy - SH
Air Force - 99

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY
Headquarters
Defense Industrial Plant Equipment Center
Memphis, Tennessee 38114-5051

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Industrial Plant Equipment Center
ATTN: DIPEC-SSM
Memphis, Tennessee 38114-5051

15

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80232B	WELDING MACHINE, ARC, AC/DC, TRANSFORMER RECTIFIER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

16

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.